CHAPTER 4:

DATA ANALYTIC APPLICATION FOR BUSINESS

(Part 2): ETL in Alteryx Designer

Modern Data Analytics Tool: ALTERYX

Use single workflow for Data Blending, Analytics and Reporting.

- A drag and drop visual workflow – no programming required.
- Seamless blending of data.
- 60+ built-in tools for spatial and R-based predictive analytics.
- Simple creation of reports, data for visualization or analytics apps.
- User productivity in hours not days.

Hands-On: ALTERYX DESIGNER

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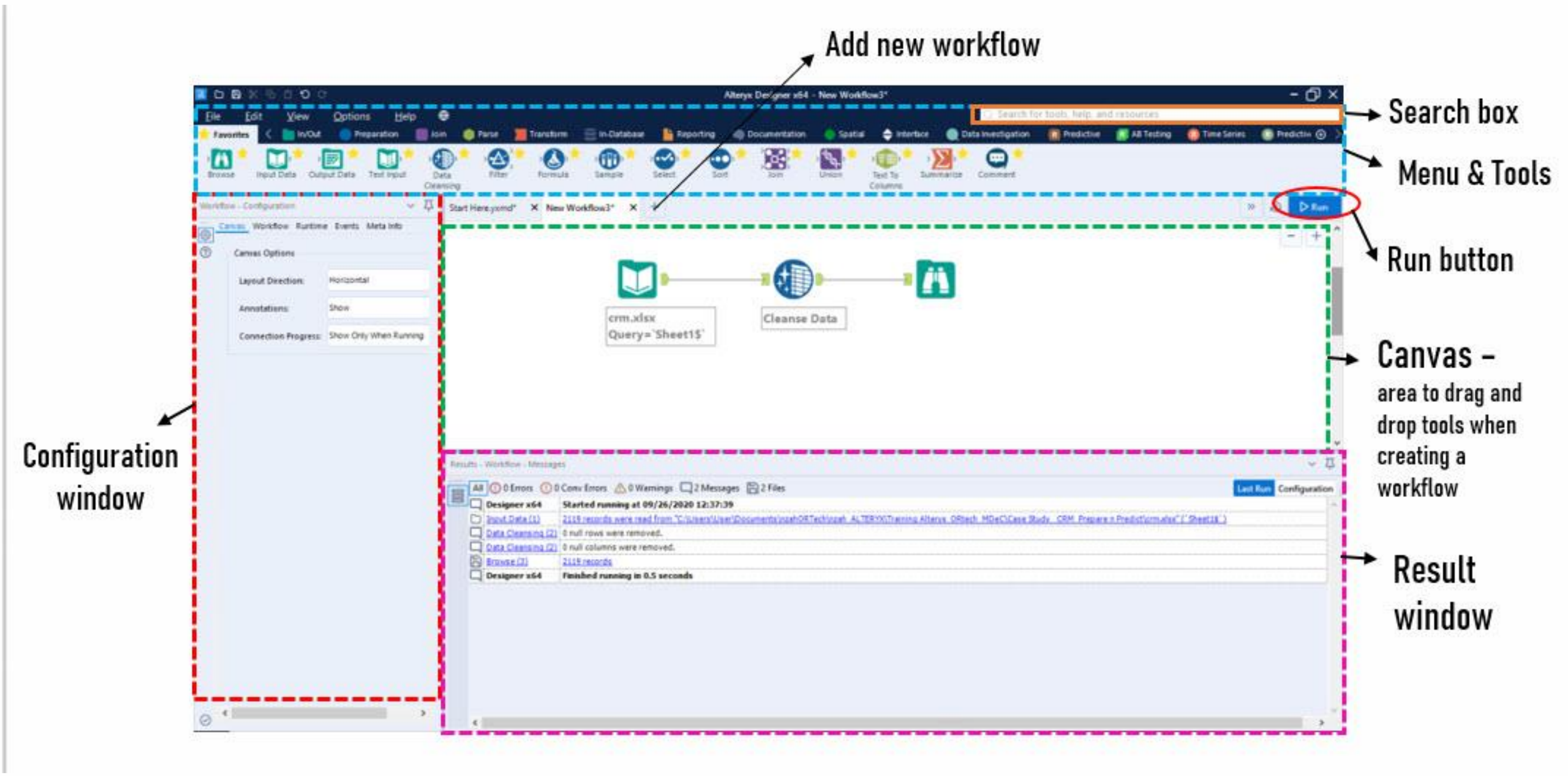


2.0: Read Input Data

Input Data Tool + **Browse Tool**

- Use **Input Data tool** to add data to your workflow by connecting it to a file or database.

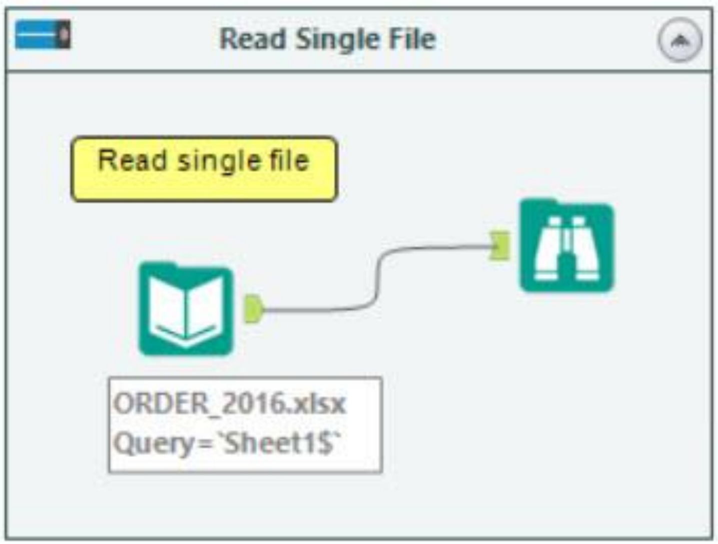
- One Input Data tool can be used to read **multiple files** using the wildcard format such as *.csv, as long as the files all contain the same number of fields, where the data types for each field are the same.

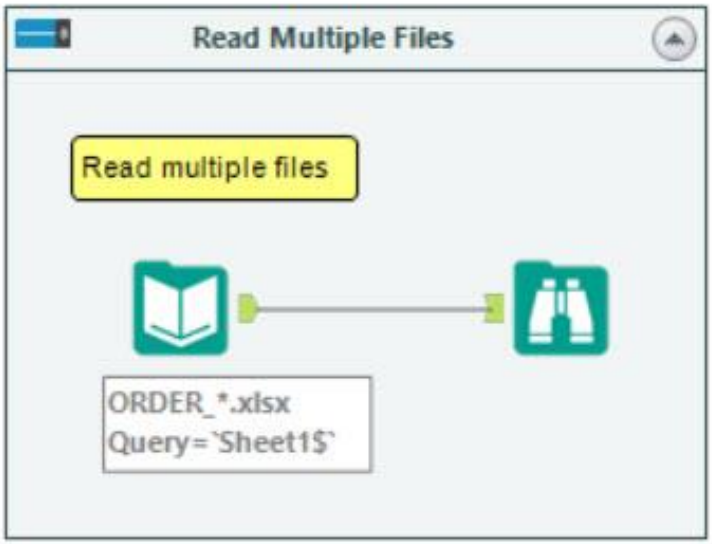
- Use **Browse tool** to view data from a connected tool. You can see data **Profile** information for multiple columns at once in a single holistic view, or for a single column of data.

- You can view information on data type, number of records, data quality, and a variety of statistics.

2.0: Read Input Data

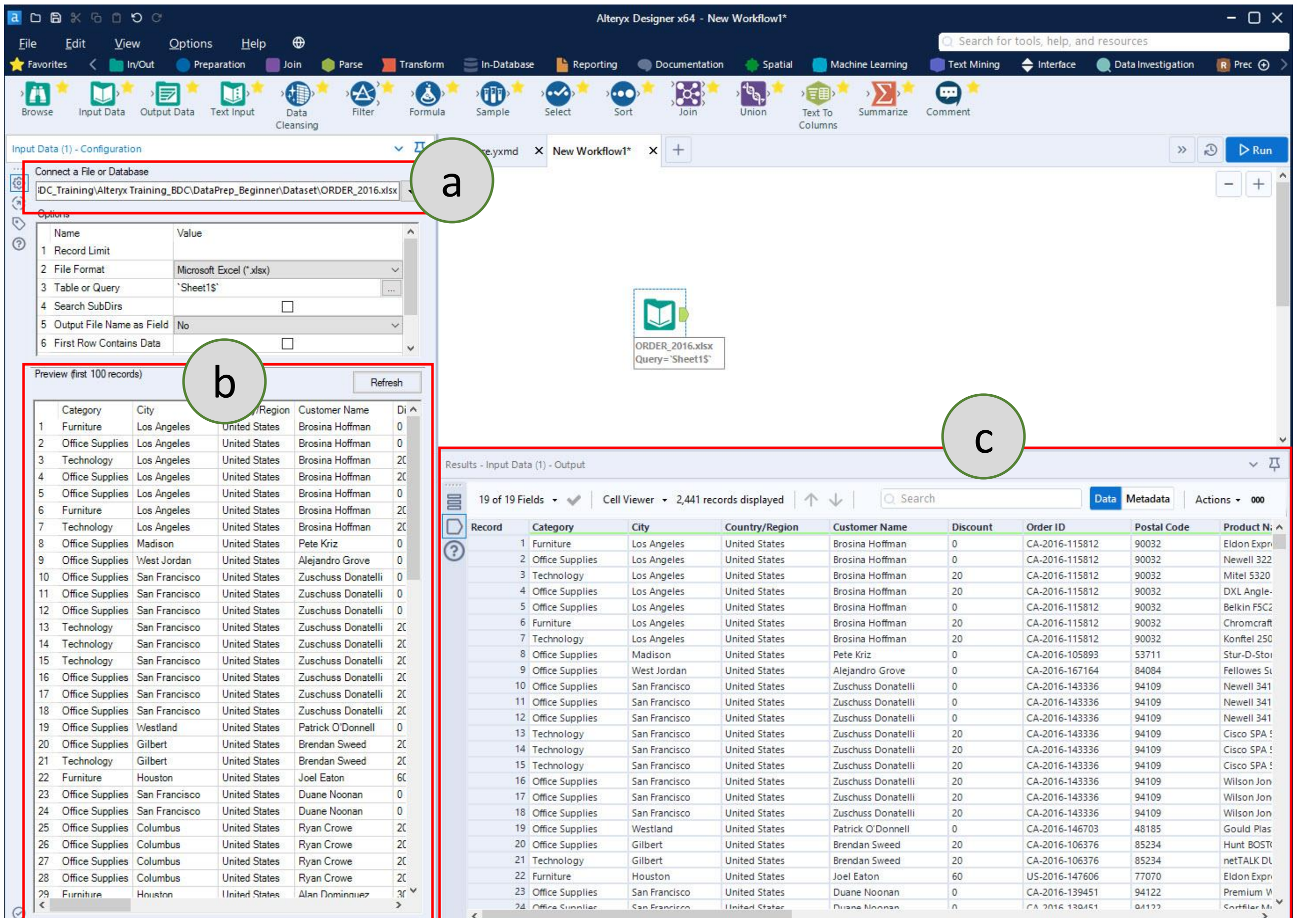
Example:





2.0: Read Input Data

Configurations:



a: Data connection

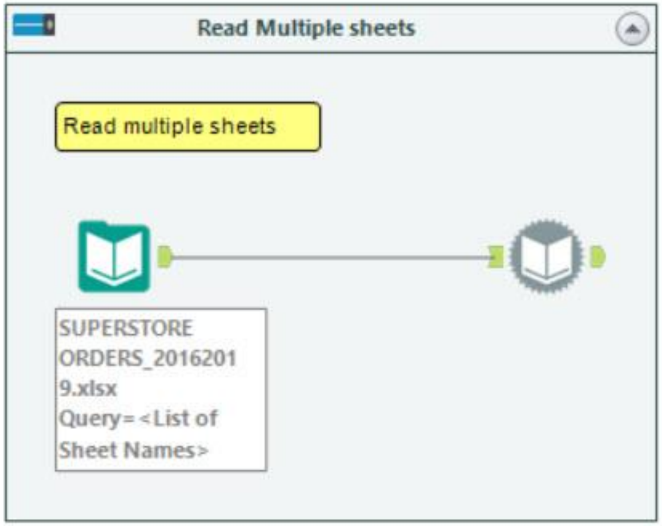
b: Data preview

c: Results

2.0: Read Input Data

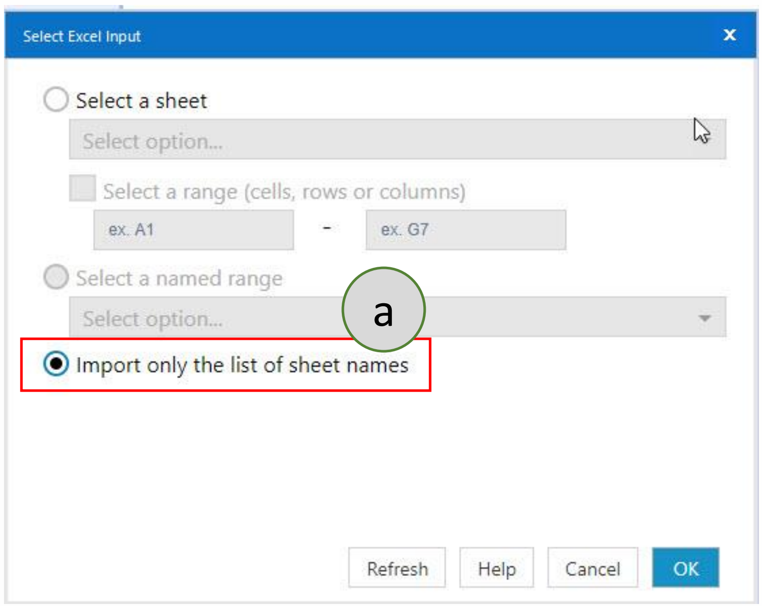
Input Data Tool + **Dynamic Input Tool**

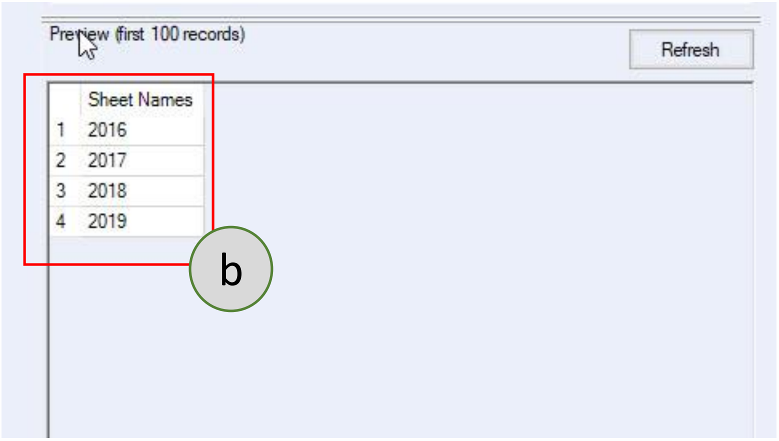
- One Input Data tool and **Dynamic Input** tool can be used to read **multiple sheets** from one Excel file.
- **Example:**



2.0: Read Input Data

Configurations – Input Data tool:



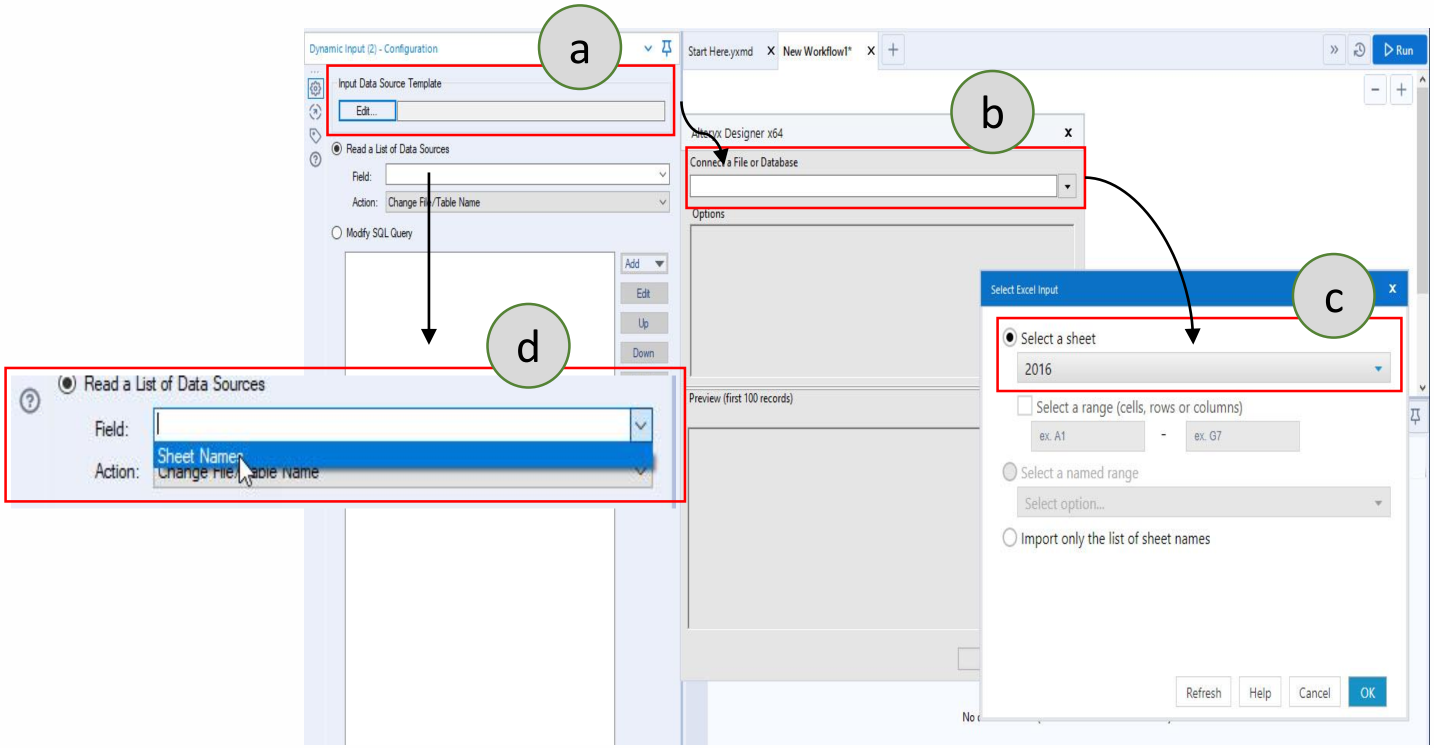


a: Configure in **Input Data** tool > Select Excel Input box

b: Only sheet names will be listed in Preview

2.0: Read Input Data

Configurations – Dynamic Input tool :



a, b: connect to input data source
c: select any (one) sheets from the data source
d: select **Sheet Names**

2.0: Read Input Data

Text Input Tool

- Use **Text Input tool** to manually enter text to create small data files for input. This can be useful in testing and creating Lookup tables while you build your workflow.

- The data input from the Text Input tool saves within the workflow, so you can share the workflow without having to provide an input data file.

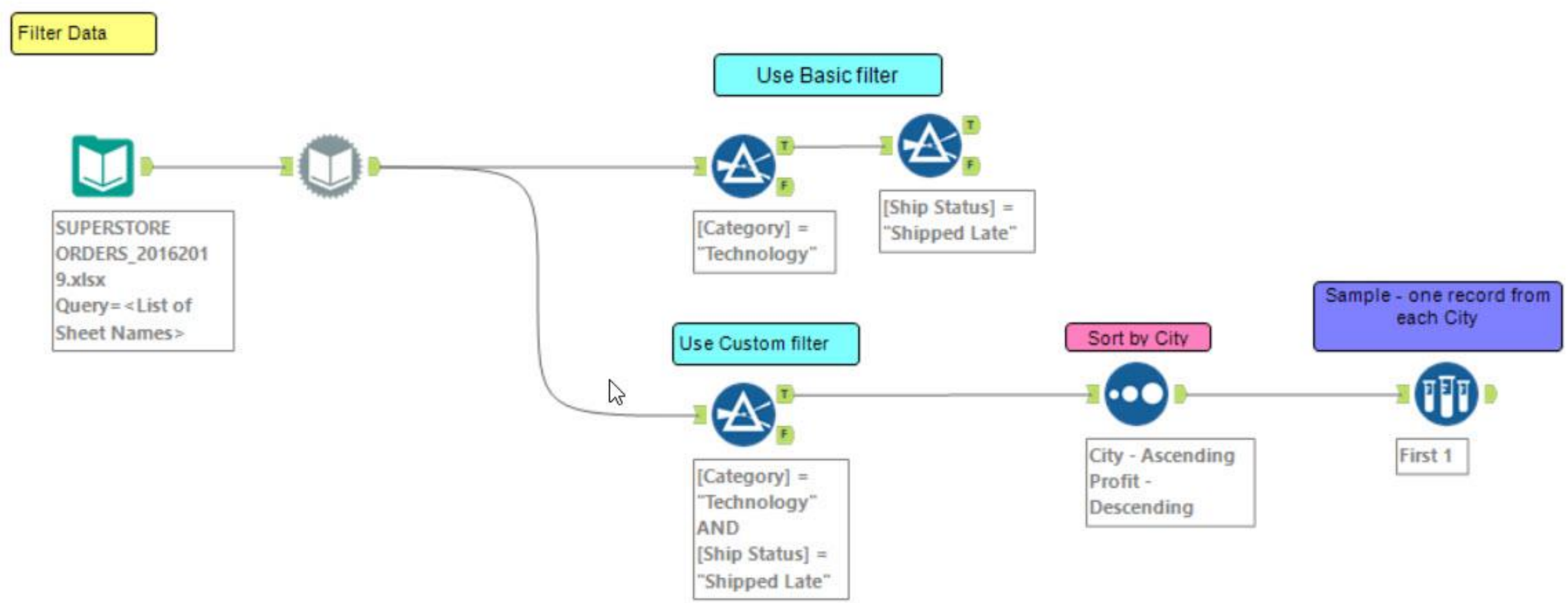
Sample

Delete Insert

	RecordID	ITEM
1	5	12 Red Patagonia
2	15	06 Purple Sierra Design
3	52	13 Yellow Mountain Hardware
4	128	00 Orange North Face
5	131	01 Frost ISIS
6	287	34 Cranberry Blisswear
7	391	45 Black OpenCasket
8	441	90 OffWhite Frozen Tundra
9	464	12 Purple Purple Haze
10	466	22 Red Moab Extreme
11	593	89 Green Rocks and Chicks
12	741	50 Orange Extreme
13	850	60 Red Adidas

3.0: Preparation

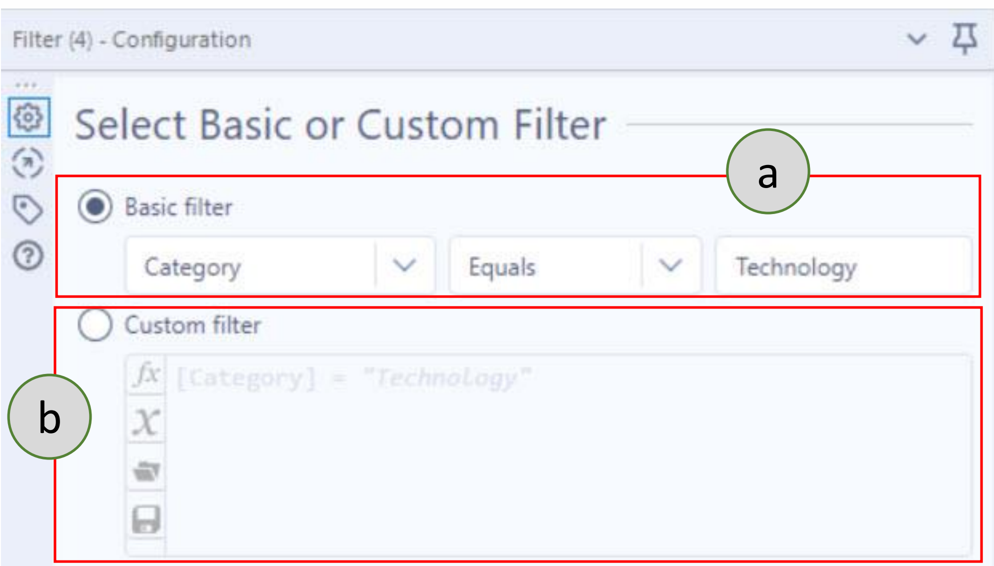
Example #1:

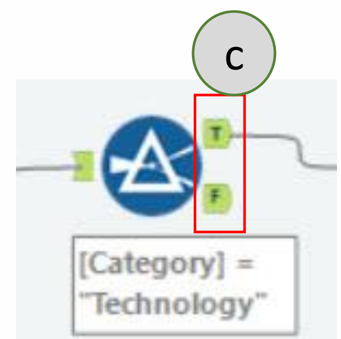


3.0: Preparation

Filter Tool

- Use **Filter** to select data using a condition. Rows of data that meet the condition are output to the **T**rue anchor. Rows of data that do not meet the condition are output to the **F**alse anchor.



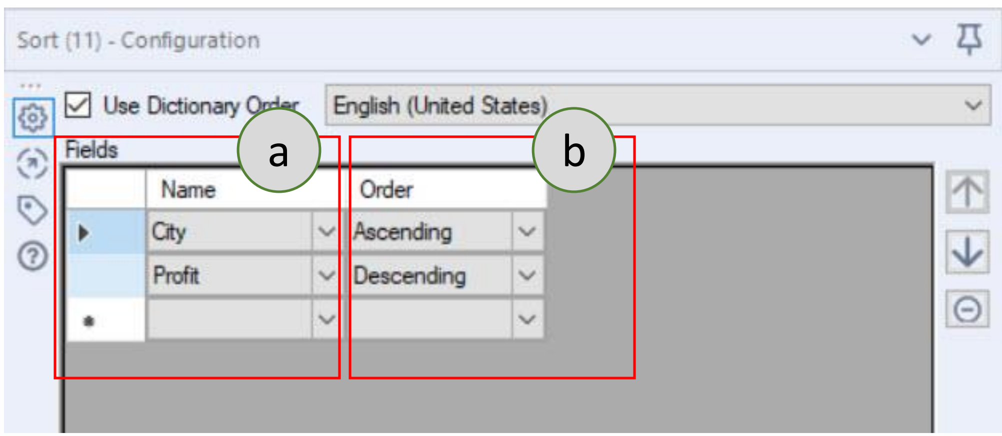


a: setting for Basic filter
b: setting for Custom filter
c: output anchor (T, F)

3.0: Preparation

Sort Tool

- Use **Sort** to arrange the records in a table in alphanumeric order based on the values of the specified data fields.



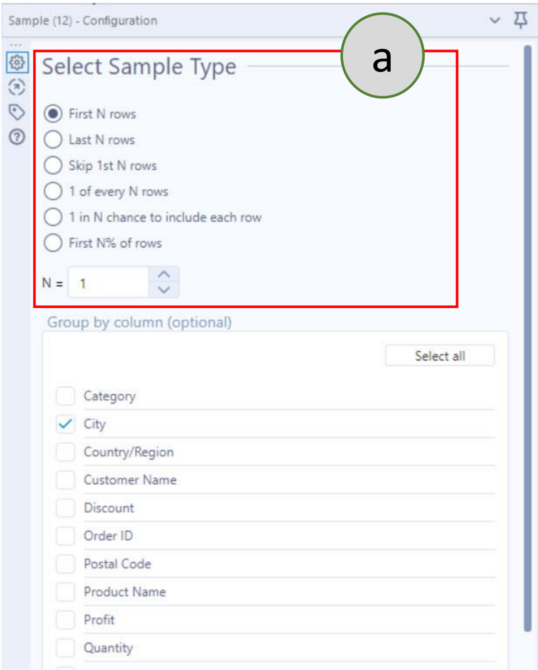
a: select field to be sorted

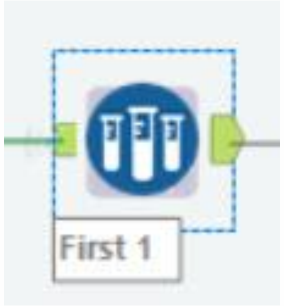
b: select rule for sorting

3.0: Preparation

Sample Tool

- Use **Sample** to limit the data stream to a specified number, percentage, or random set of rows. In addition, the Sample tool applies the selected configuration to the columns selected to group by.



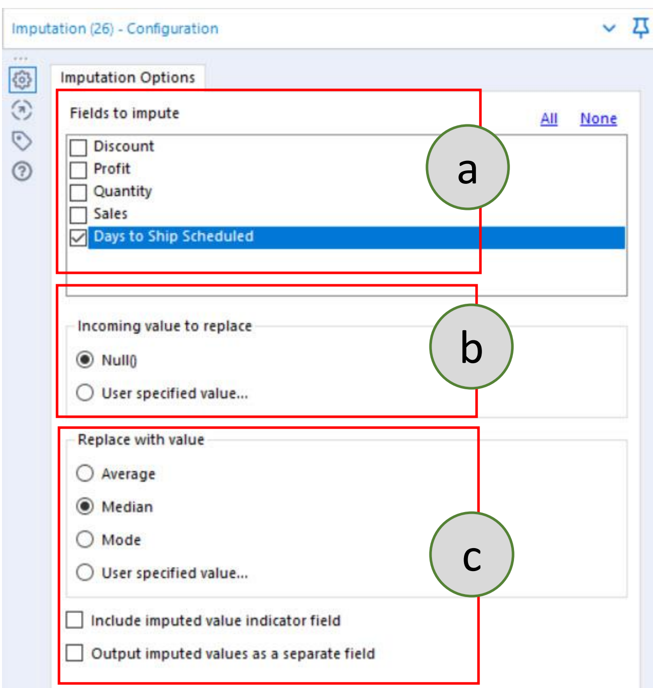


a: select one sample type

3.0: Preparation

Imputation Tool

- Use **Imputation** to replace a specified value within one or more numeric data fields with another specified value.
- A typical use case is to replace all NULL() values with the average of the remaining values for particular field.



a: select field to impute

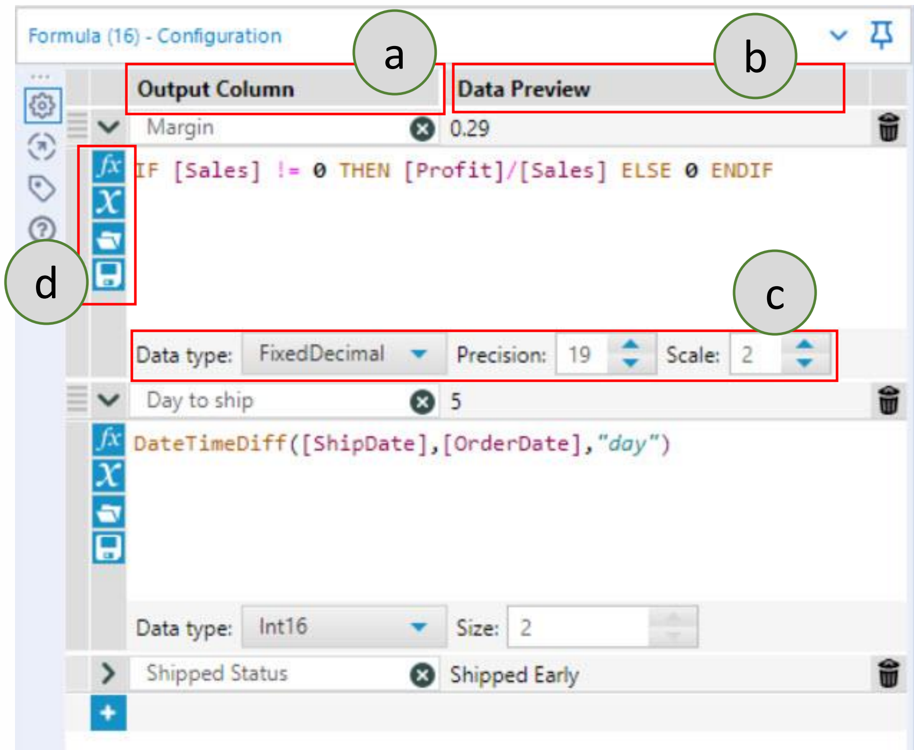
b: select the incoming value that require imputation

c: value to be replaced/impute

3.0: Preparation

Formula Tool

- Use **Formula** to create new columns, update columns, and use one or more expressions to perform a variety of calculations and operations.



a: Select column to process/apply
b: Data preview
c: Setting output data type after process
d: Functions, variables, saved expression

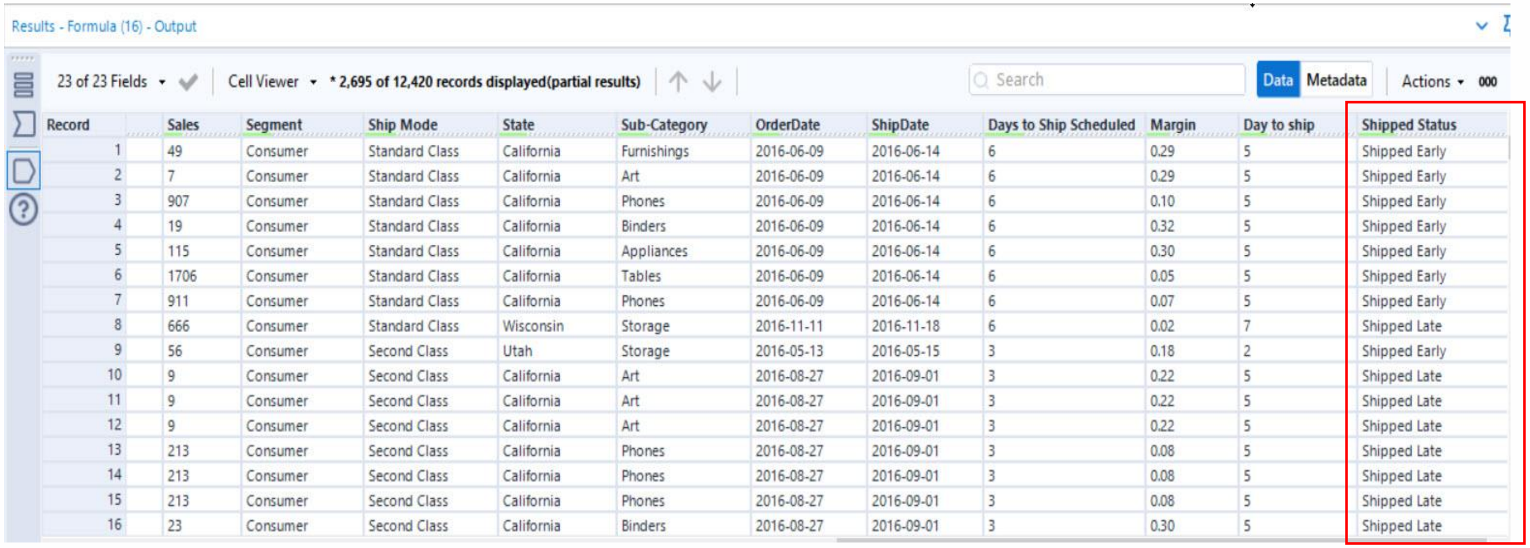
Exercise #1

With the same workflow, create a new output column to add info about shipping status with the following conditions:

If [Day to ship] = [Day to shipped scheduled] then “Shipped On Time”
If [Day to ship] > [Day to shipped scheduled] then “Shipped Late”
If [Day to ship] < [Day to shipped scheduled] then “Shipped Early”

Hint: Add new output column on the same Formula tool.

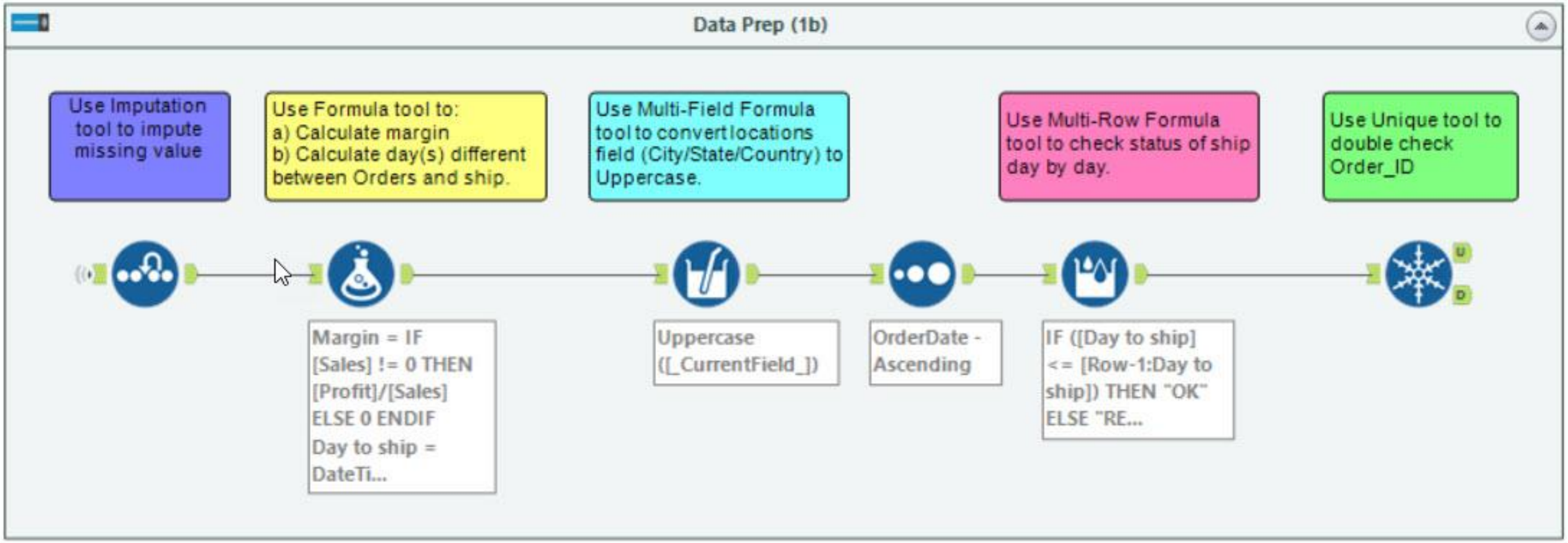
Sample result



Record	Sales	Segment	Ship Mode	State	Sub-Category	OrderDate	ShipDate	Days to Ship Scheduled	Margin	Day to ship	Shipped Status
1	49	Consumer	Standard Class	California	Furnishings	2016-06-09	2016-06-14	6	0.29	5	Shipped Early
2	7	Consumer	Standard Class	California	Art	2016-06-09	2016-06-14	6	0.29	5	Shipped Early
3	907	Consumer	Standard Class	California	Phones	2016-06-09	2016-06-14	6	0.10	5	Shipped Early
4	19	Consumer	Standard Class	California	Binders	2016-06-09	2016-06-14	6	0.32	5	Shipped Early
5	115	Consumer	Standard Class	California	Appliances	2016-06-09	2016-06-14	6	0.30	5	Shipped Early
6	1706	Consumer	Standard Class	California	Tables	2016-06-09	2016-06-14	6	0.05	5	Shipped Early
7	911	Consumer	Standard Class	California	Phones	2016-06-09	2016-06-14	6	0.07	5	Shipped Early
8	666	Consumer	Standard Class	Wisconsin	Storage	2016-11-11	2016-11-18	6	0.02	7	Shipped Late
9	56	Consumer	Second Class	Utah	Storage	2016-05-13	2016-05-15	3	0.18	2	Shipped Early
10	9	Consumer	Second Class	California	Art	2016-08-27	2016-09-01	3	0.22	5	Shipped Late
11	9	Consumer	Second Class	California	Art	2016-08-27	2016-09-01	3	0.22	5	Shipped Late
12	9	Consumer	Second Class	California	Art	2016-08-27	2016-09-01	3	0.22	5	Shipped Late
13	213	Consumer	Second Class	California	Phones	2016-08-27	2016-09-01	3	0.08	5	Shipped Late
14	213	Consumer	Second Class	California	Phones	2016-08-27	2016-09-01	3	0.08	5	Shipped Late
15	213	Consumer	Second Class	California	Phones	2016-08-27	2016-09-01	3	0.08	5	Shipped Late
16	23	Consumer	Second Class	California	Binders	2016-08-27	2016-09-01	3	0.30	5	Shipped Late

3.0: Preparation

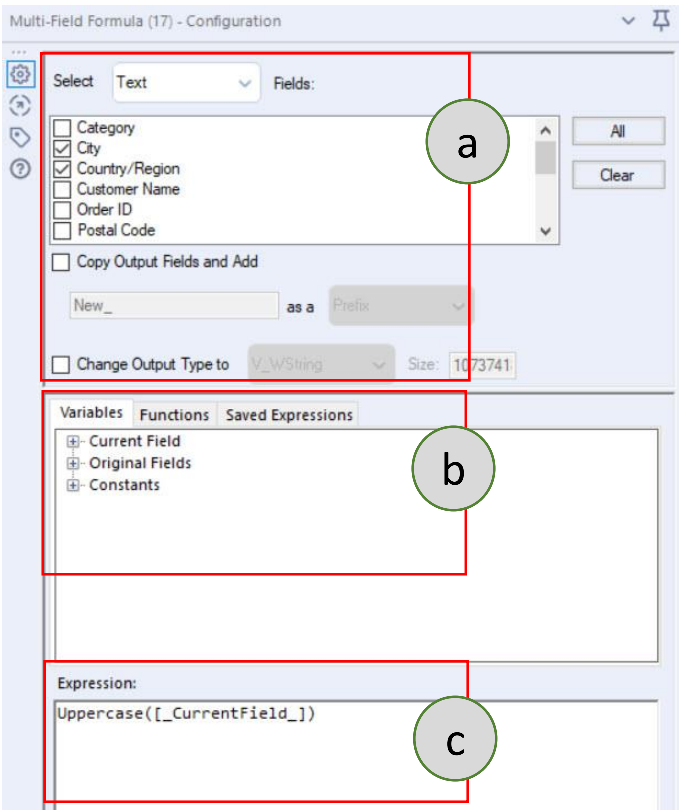
Example #2:

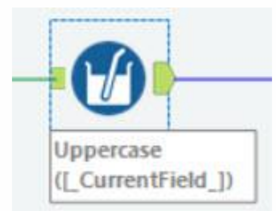


3.0: Preparation

Multi-Field Formula Tool

- Use **Multi-Field Formula** tool to create or update multiple fields using a single expression.



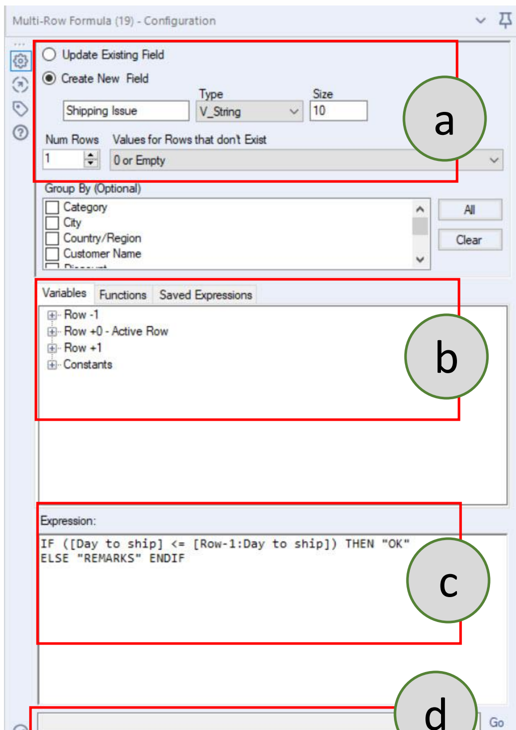


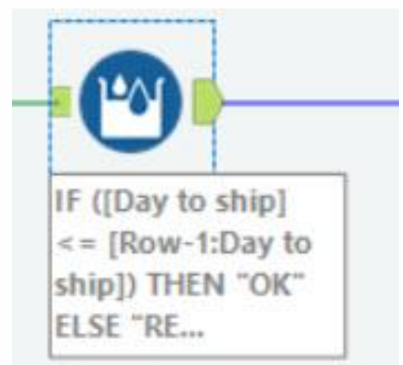
a: Select column to process/apply
b: Select variable and function
c: Write expression

3.0: Preparation

Multi-Row Formula Tool

- Use **Multi-Row Formula** tool to take the concept of the Formula tool a step further. This tool allows you to utilize row data as part of the formula creation and is useful for parsing complex data, and creating running totals, averages, percentages, and other mathematical calculations. This tool can only update one field per tool instance.



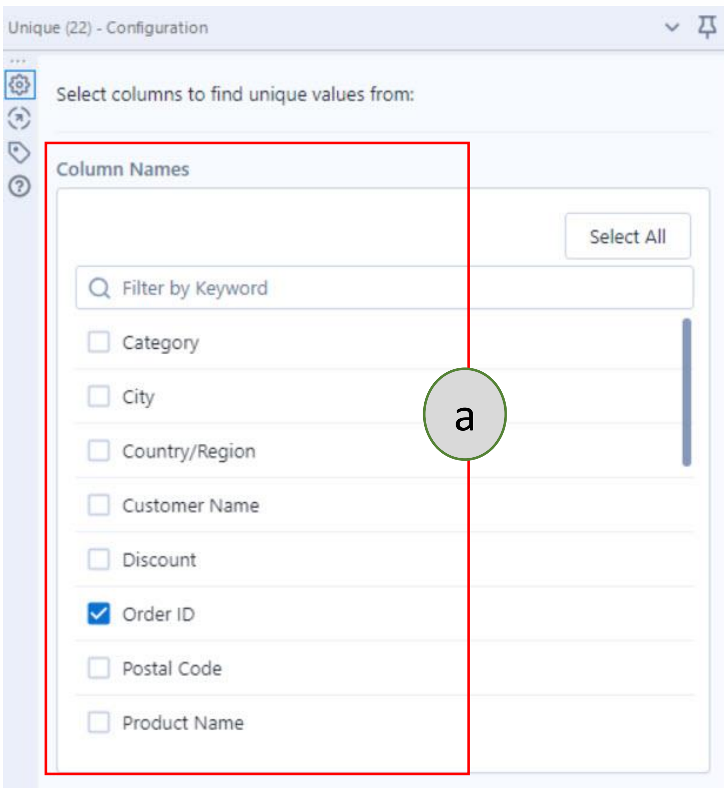


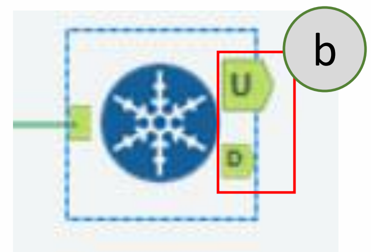
a: Select or create field/column to process
b: Select variable and function
c: Write expression
d: message if there is malfunction in the expression

3.0: Preparation

Unique Tool

- Use **Unique** tool to distinguish whether a data record is unique or a duplicate by grouping on one or more specified fields, then sorting on those fields.





a: Select field/column to process
b: Output anchor

Exercise #2

The data set GiftRecords.xlsx contains a yearly records (2010-2015 by sheets) about donations. Find the highest Gift Amounts by each College for San Francisco City.

Sample solution:

Results - Sample (7) - Output

10 of 10 Fields | Cell Viewer | 12 records displayed | Search | Data | Metadata | Actions

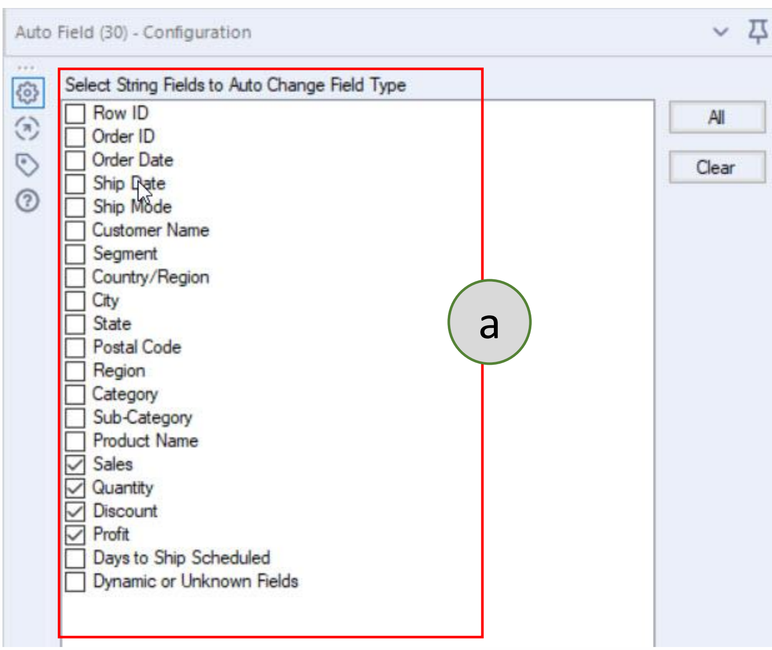
Record	Allocation Subcategory	City	College	Gift Allocation	Gift Amount
1	Trustees Fund	San Francisco	College of Agriculture and Natural Resources	Endowment	14961
2	College of Arts and Sciences	San Francisco	College of Arts and Sciences	Scholarship	14156
3	Minority Scholarship Fund	San Francisco	College of Business	Scholarship	8877
4	College of Social Science	San Francisco	College of Communication Arts and Sciences	Scholarship	2307
5	College of Education	San Francisco	College of Education	Scholarship	24932
6	University Annual Fund	San Francisco	College of Engineering	Endowment	23796
7	College of Arts and Sciences	San Francisco	College of Music	Scholarship	13836
8	College of Agriculture and Natural Resources	San Francisco	College of Natural Science	Scholarship	14979
9	University Annual Fund	San Francisco	College of Nursing	Endowment	7094
10	Trustees Fund	San Francisco	College of Political Science	Endowment	13847
11	College of Political Science	San Francisco	College of Social Science	Scholarship	21136
12	Diversity Fund	San Francisco	College of Veterinary Medicine	Endowment	2268

3.0: Preparation

Auto Field Tool

- Use **Auto Field** tool to read through all of the records of an input and set the field type to the smallest possible size relative to the data contained within the column.

- The tool correctly assigns a numeric field to a string data type where any record starts with zero and not a number.

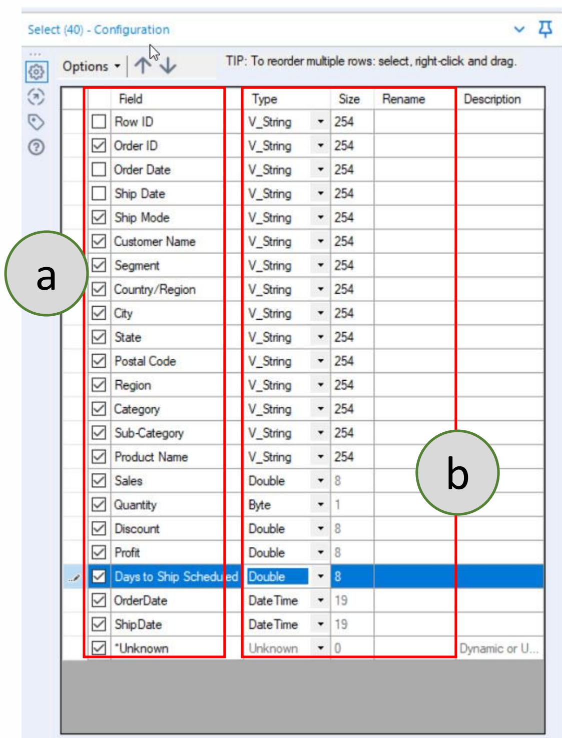


a: Select field/column to process

3.0: Preparation

Select Tool

- Another option to edit the data type, we can use **Select** tool. Also, this tool can be used to rename field, change size, adding description and rearrange/sort the order of the field.

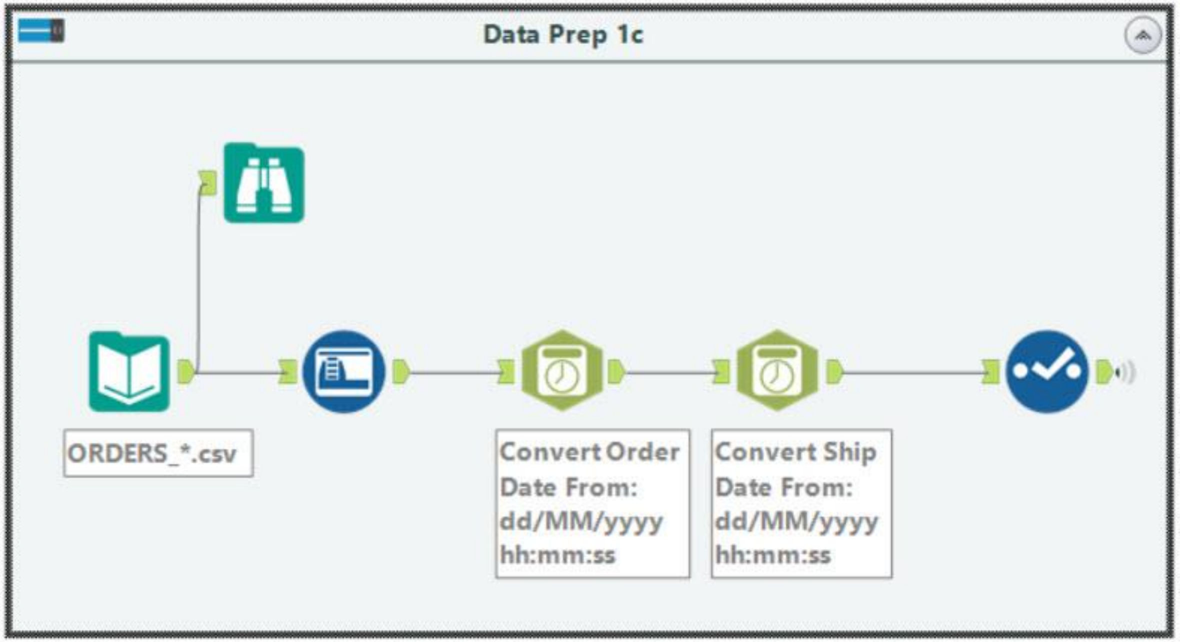


a: Tick to select field/column to process

b: Setting data type, size and rename (optional)

4.0: Parse

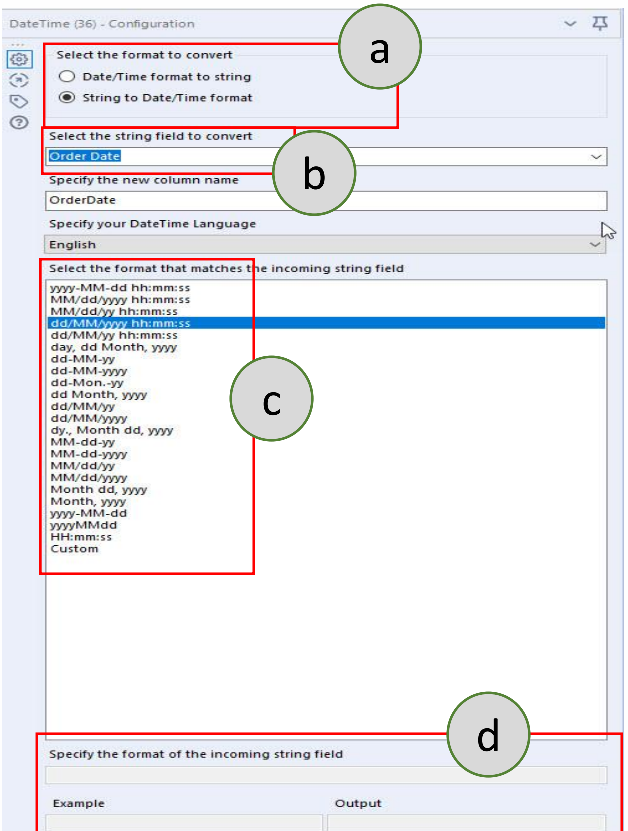
Example:



4.0: Parse

DateTime Tool

- Use **DateTime** tool to transform date-time data to and from variety of formats, including both expression-friendly and human-readable formats.



a: Tick to select format to convert

b: Select the field to process

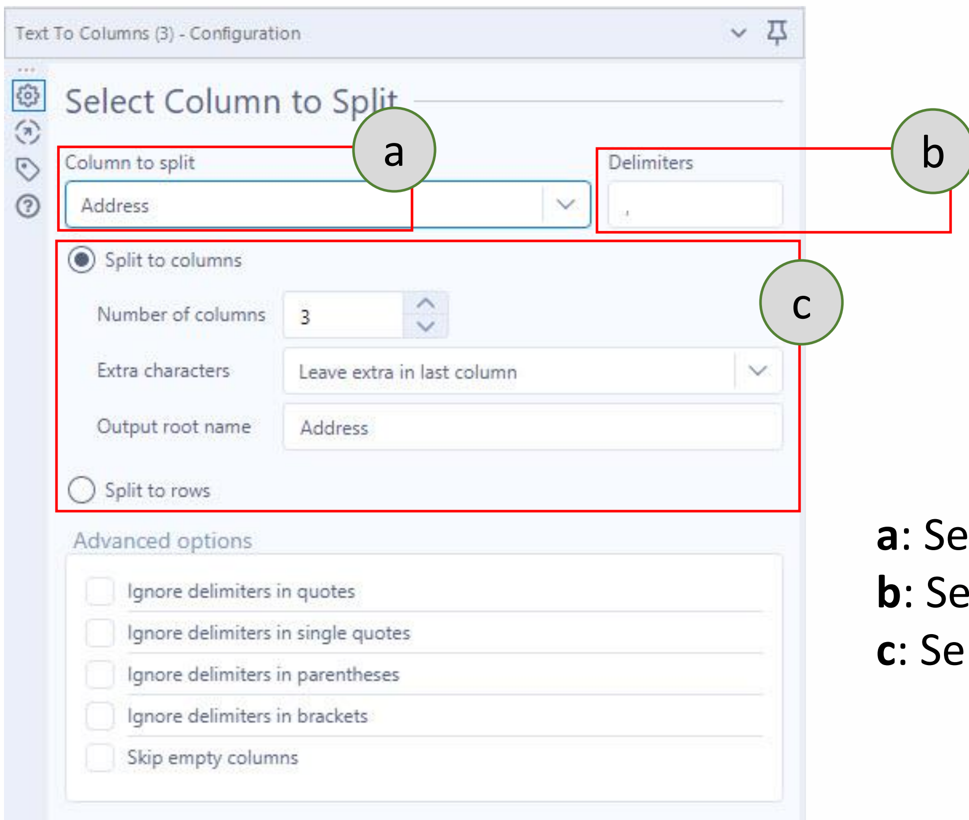
c: Select format for incoming string field (as choose in **b**)

d: Specify format for incoming string field (as choose in **b**) (optional – if choose **Custom** in **c**)

4.0: Parse

Text To Columns Tool

- Use **Text To Columns** tool to take the text in one column and split the string value into separate, multiple columns (or rows), based on a single or multiple delimiters.



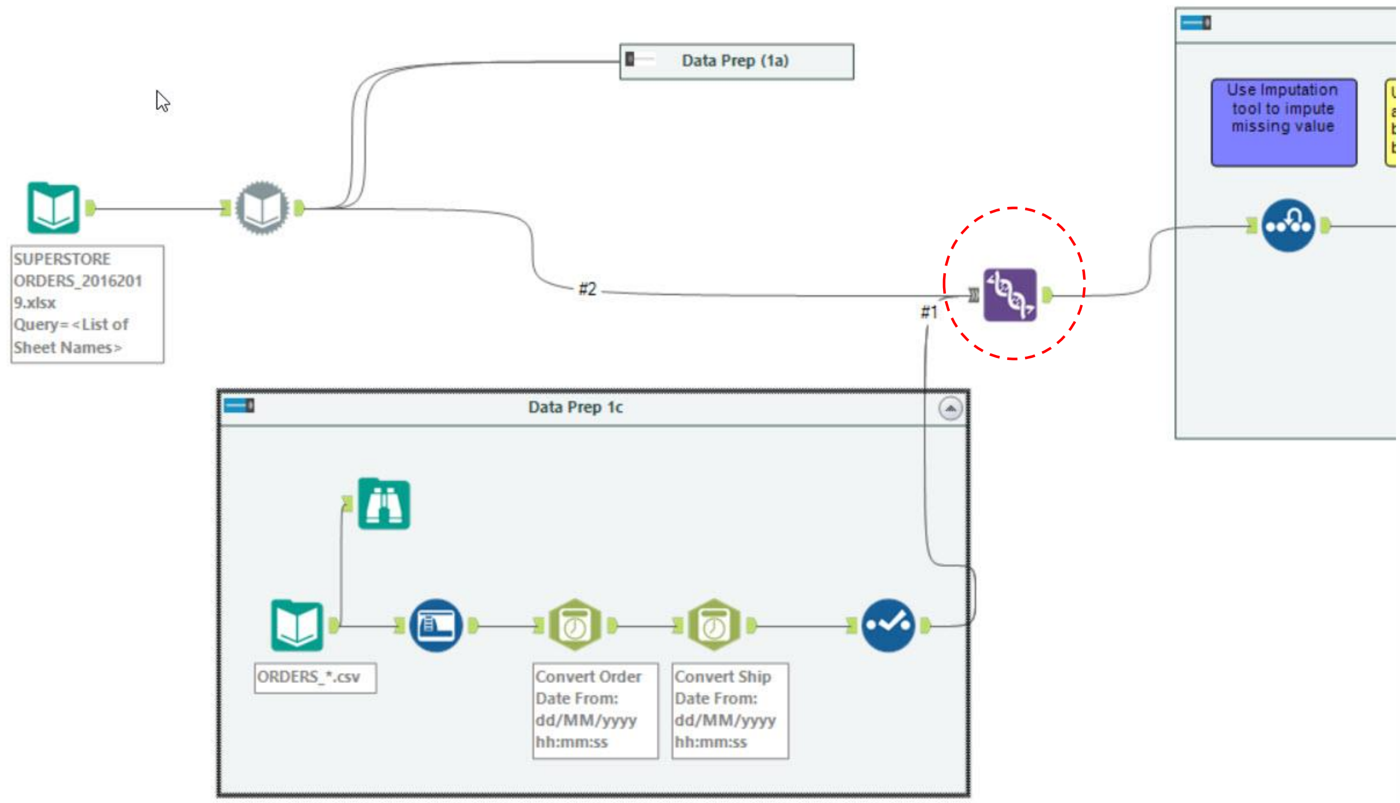
a: Select field/column to process

b: Set delimiter use in the data source

c: Select either to Split to columns or row

5.0: Join

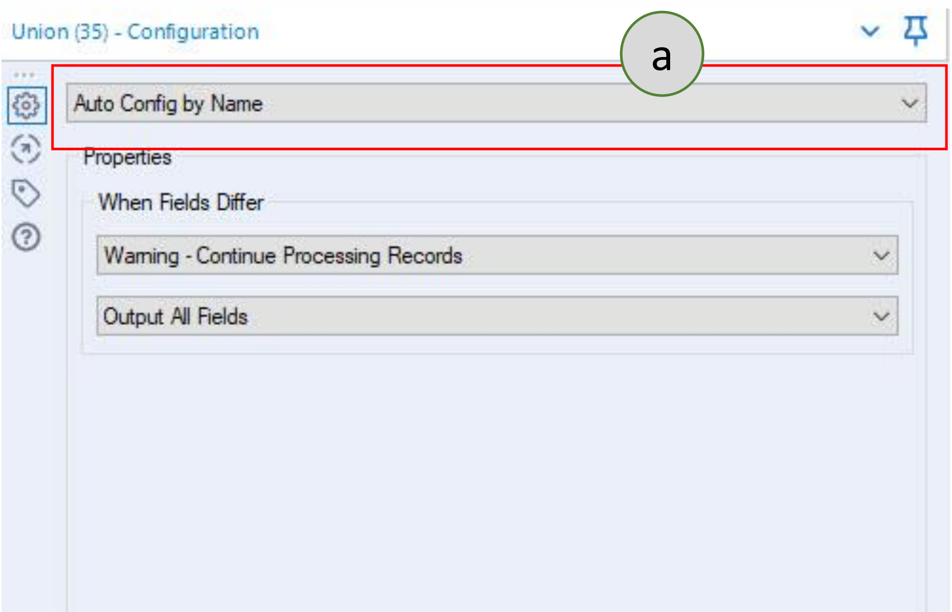
Example:

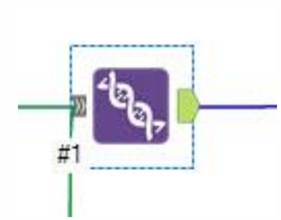


5.0: Join

Union Tool

- Use **Union** tool to combine 2 or more datasets on column names or position. In the output, each column contains the rows from each input. You can configure how the columns stack or match up in the output.



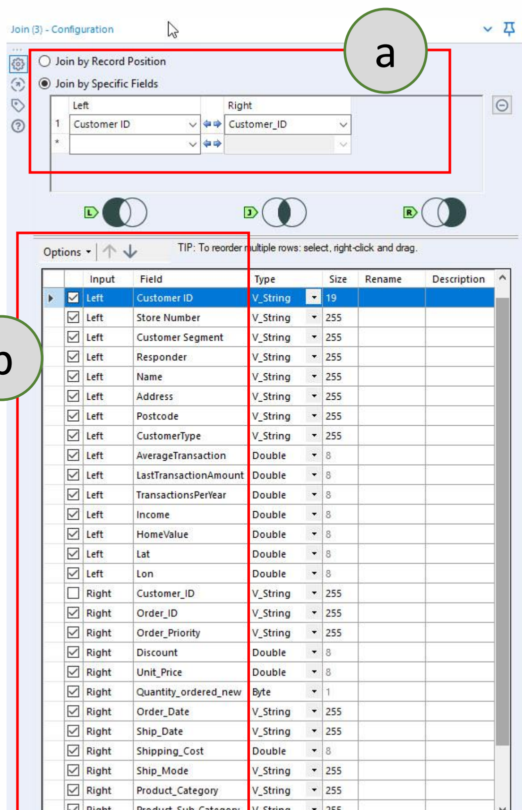


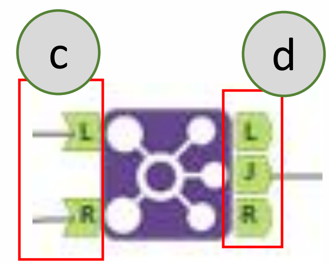
a: Select one rule for union as either
- *Auto Config by Name,* or
- *Auto Config by Position,* or
- *Manually Configure Fields*

5.0: Join

Join Tool

- Use **Join** tool to combine 2 inputs based on common fields between the 2 tables. You can also Join 2 data streams based on record position.





a: Select rule for join.

b: Tick fields to be joined.

c: Connect the tables to be joined - one from **L**eft, and one from **R**ight.

d: Output after join process –

L = all records from Left that do not match to join.

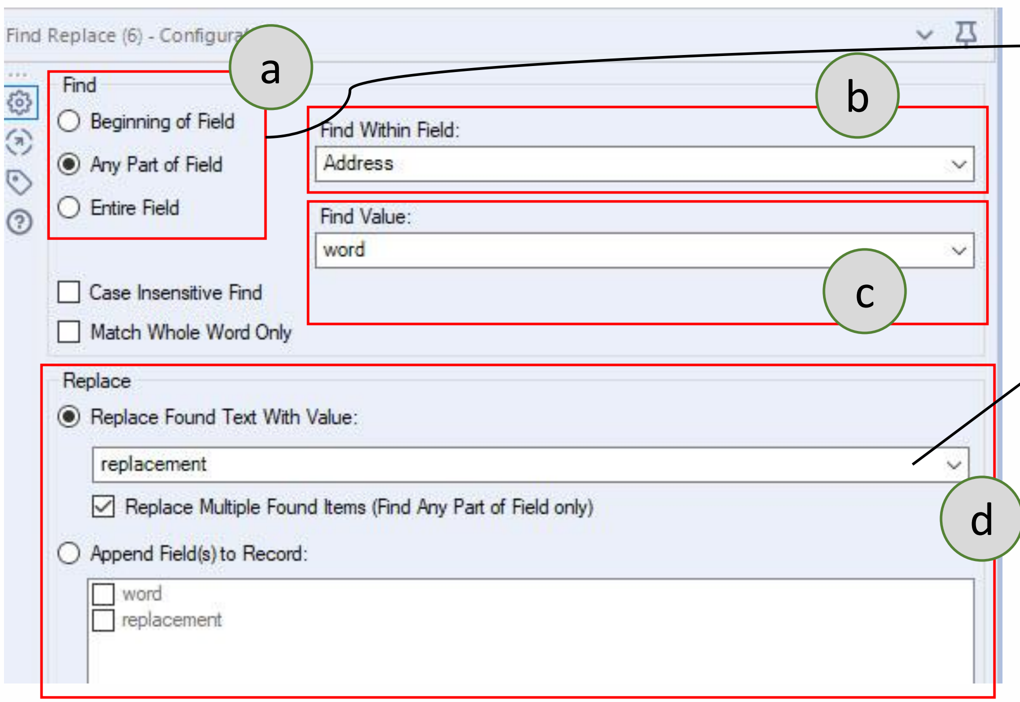
J = all records that successfully joined.

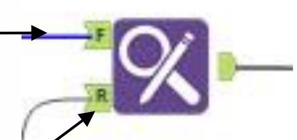
R = all records from Right that do not match to join.

5.0: Join

Find Replace Tool

- Use **Find Replace** tool find a string in one column of a dataset and look up and replace it with the specified value from another dataset.



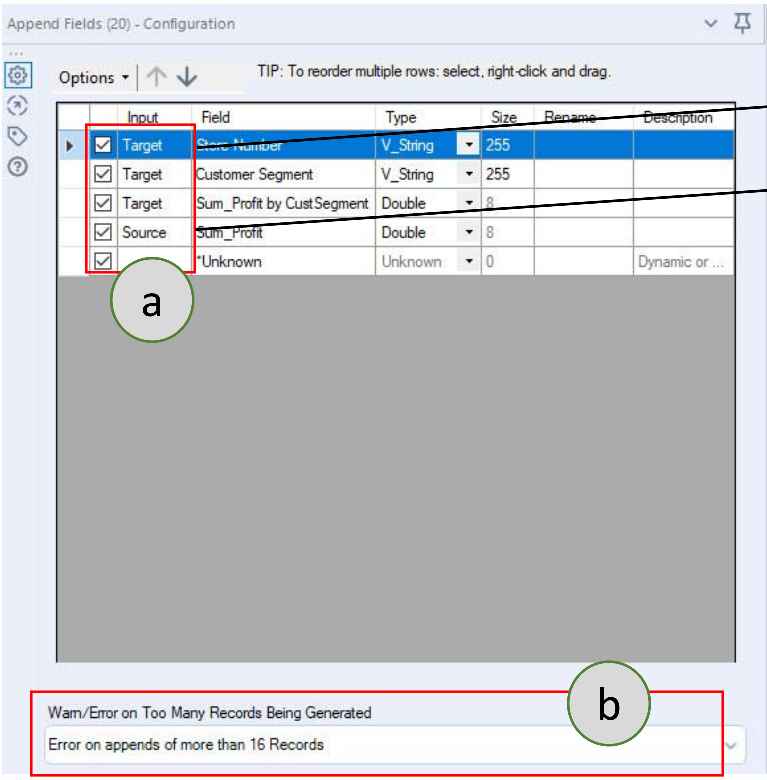


a: Select part to be replaced.
b: Select field to process.
c: Select field that keep the list of words/values to be replaced.
d: Select field that keep the list of replacement word/value.

5.0: Join

Append Fields Tool

- Use **Append Fields** tool to append the fields of one small input (Source) to every record of another larger input (Target). The result is a Cartesian join. In a Cartesian join, every row from one table is joined to every row of another table.



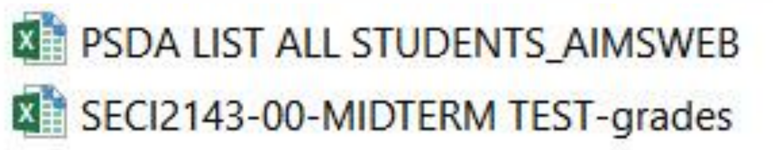
a: Target field that will be appended with record from source field.

b: Warn/Error message

Note: The append tool performs a Cartesian Join on the two input datasets, so if both datasets have 100 rows, the output will have 10,000 rows. The data can quickly become very large, and so it is recommended to leave the error setting on to prevent this from happening.

Exercise #3

Prep the following data to generate new records contains the list of students by section (SEC) in separate sheets in MS Excel. Also, identify how many students do not take the midterm test.



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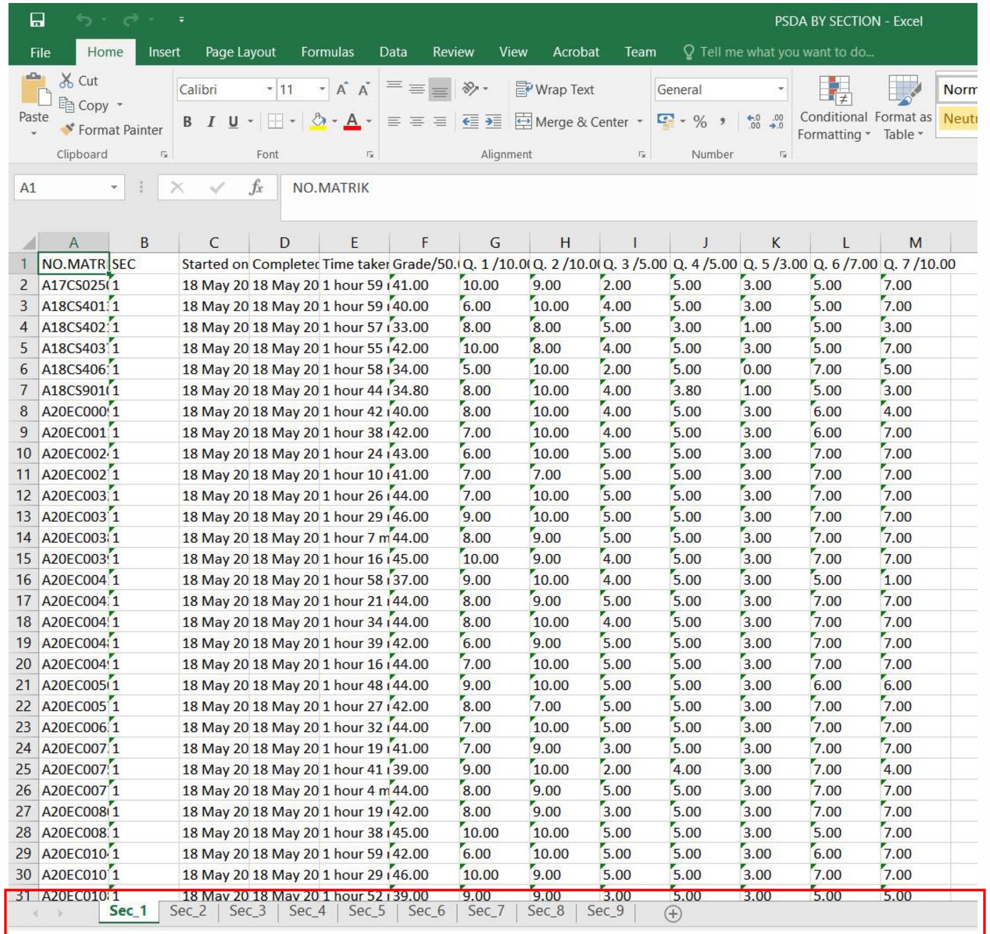
Exercise #3

Sample solution:

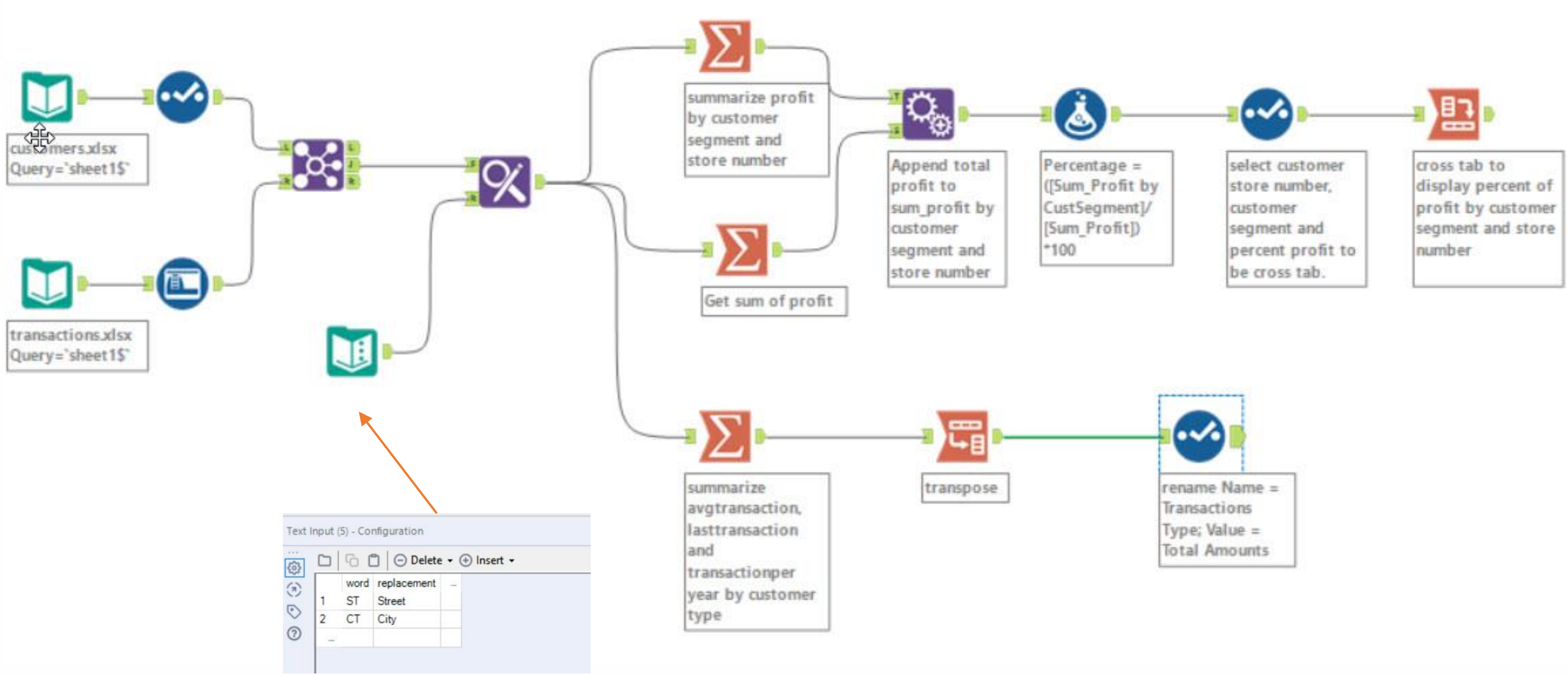


	A	B	C	D	E	F	G	H	I	J	K	L	M
1	NO.MATR	SEC	Started on	Completed	Time taken	Grade/50.	Q. 1 /10.0	Q. 2 /10.0	Q. 3 /5.00	Q. 4 /5.00	Q. 5 /3.00	Q. 6 /7.00	Q. 7 /10.00
2	A17CS025	1	18 May 20	18 May 20	1 hour 59	41.00	10.00	9.00	2.00	5.00	3.00	5.00	7.00
3	A18CS401	1	18 May 20	18 May 20	1 hour 59	40.00	6.00	10.00	4.00	5.00	3.00	5.00	7.00
4	A18CS402	1	18 May 20	18 May 20	1 hour 57	33.00	8.00	8.00	5.00	3.00	1.00	5.00	3.00
5	A18CS403	1	18 May 20	18 May 20	1 hour 55	42.00	10.00	8.00	4.00	5.00	3.00	5.00	7.00
6	A18CS406	1	18 May 20	18 May 20	1 hour 58	34.00	5.00	10.00	2.00	5.00	0.00	7.00	5.00
7	A18CS901	1	18 May 20	18 May 20	1 hour 44	34.80	8.00	10.00	4.00	3.80	1.00	5.00	3.00
8	A20EC000	1	18 May 20	18 May 20	1 hour 42	40.00	8.00	10.00	4.00	5.00	3.00	6.00	4.00
9	A20EC001	1	18 May 20	18 May 20	1 hour 38	42.00	7.00	10.00	4.00	5.00	3.00	6.00	7.00
10	A20EC002	1	18 May 20	18 May 20	1 hour 24	43.00	6.00	10.00	5.00	5.00	3.00	7.00	7.00
11	A20EC002	1	18 May 20	18 May 20	1 hour 10	41.00	7.00	7.00	5.00	5.00	3.00	7.00	7.00
12	A20EC003	1	18 May 20	18 May 20	1 hour 26	44.00	7.00	10.00	5.00	5.00	3.00	7.00	7.00
13	A20EC003	1	18 May 20	18 May 20	1 hour 29	46.00	9.00	10.00	5.00	5.00	3.00	7.00	7.00
14	A20EC003	1	18 May 20	18 May 20	1 hour 7 m	44.00	8.00	9.00	5.00	5.00	3.00	7.00	7.00
15	A20EC003	1	18 May 20	18 May 20	1 hour 16	45.00	10.00	9.00	4.00	5.00	3.00	7.00	7.00
16	A20EC004	1	18 May 20	18 May 20	1 hour 58	37.00	9.00	10.00	4.00	5.00	3.00	5.00	1.00
17	A20EC004	1	18 May 20	18 May 20	1 hour 21	44.00	8.00	9.00	5.00	5.00	3.00	7.00	7.00
18	A20EC004	1	18 May 20	18 May 20	1 hour 34	44.00	8.00	10.00	4.00	5.00	3.00	7.00	7.00
19	A20EC004	1	18 May 20	18 May 20	1 hour 39	42.00	6.00	9.00	5.00	5.00	3.00	7.00	7.00
20	A20EC004	1	18 May 20	18 May 20	1 hour 16	44.00	7.00	10.00	5.00	5.00	3.00	7.00	7.00
21	A20EC005	1	18 May 20	18 May 20	1 hour 48	44.00	9.00	10.00	5.00	5.00	3.00	6.00	6.00
22	A20EC005	1	18 May 20	18 May 20	1 hour 27	42.00	8.00	7.00	5.00	5.00	3.00	7.00	7.00
23	A20EC006	1	18 May 20	18 May 20	1 hour 32	44.00	7.00	10.00	5.00	5.00	3.00	7.00	7.00
24	A20EC007	1	18 May 20	18 May 20	1 hour 19	41.00	7.00	9.00	3.00	5.00	3.00	7.00	7.00
25	A20EC007	1	18 May 20	18 May 20	1 hour 41	39.00	9.00	10.00	2.00	4.00	3.00	7.00	4.00
26	A20EC007	1	18 May 20	18 May 20	1 hour 4 m	44.00	8.00	9.00	5.00	5.00	3.00	7.00	7.00
27	A20EC008	1	18 May 20	18 May 20	1 hour 19	42.00	8.00	9.00	3.00	5.00	3.00	7.00	7.00
28	A20EC008	1	18 May 20	18 May 20	1 hour 38	45.00	10.00	10.00	5.00	5.00	3.00	5.00	7.00
29	A20EC010	1	18 May 20	18 May 20	1 hour 59	42.00	6.00	10.00	5.00	5.00	3.00	6.00	7.00
30	A20EC010	1	18 May 20	18 May 20	1 hour 29	46.00	10.00	9.00	5.00	5.00	3.00	7.00	7.00
31	A20EC010	1	18 May 20	18 May 20	1 hour 52	39.00	9.00	9.00	3.00	5.00	3.00	5.00	5.00

9 sections

6.0: Transform

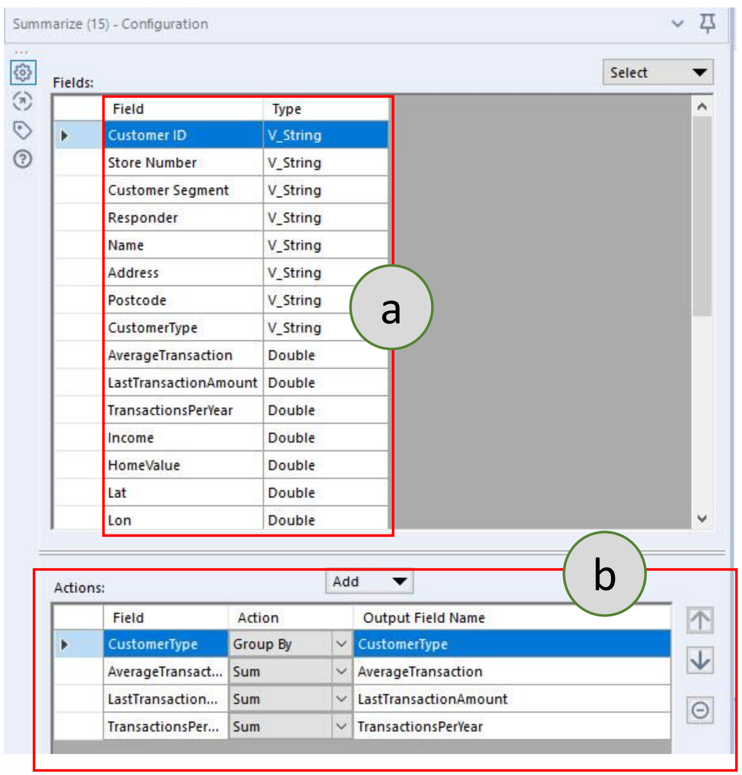
Example:



6.0: Transform

Summarize Tool

- Use **Summarize** tool to perform various actions (functions and calculations) on your data.

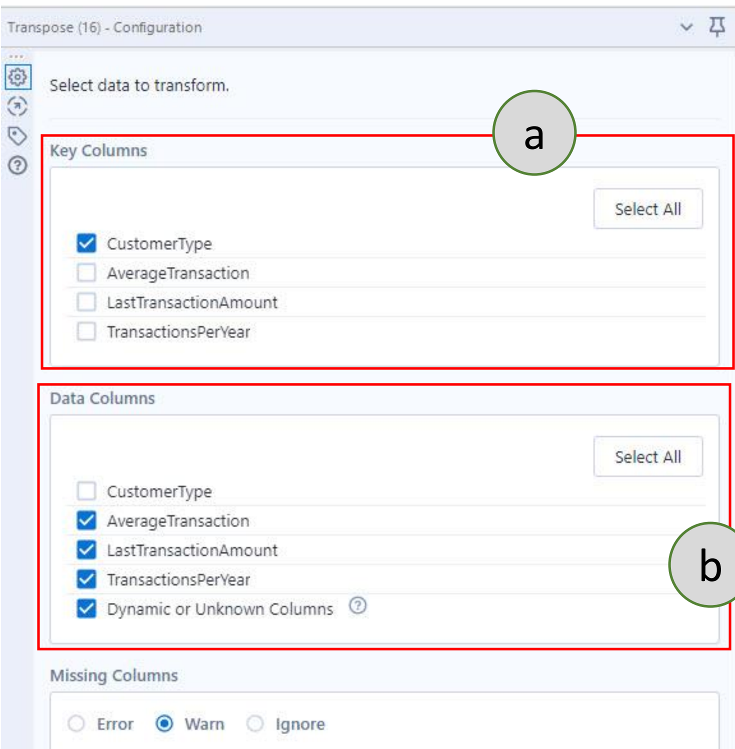


a: Select field to process.
b: Select action to be done.

6.0: Transform

Transpose Tool

- Use **Transpose** tool to pivot the orientation of the data table. This tool is the reverse application of the Cross Tab tool.



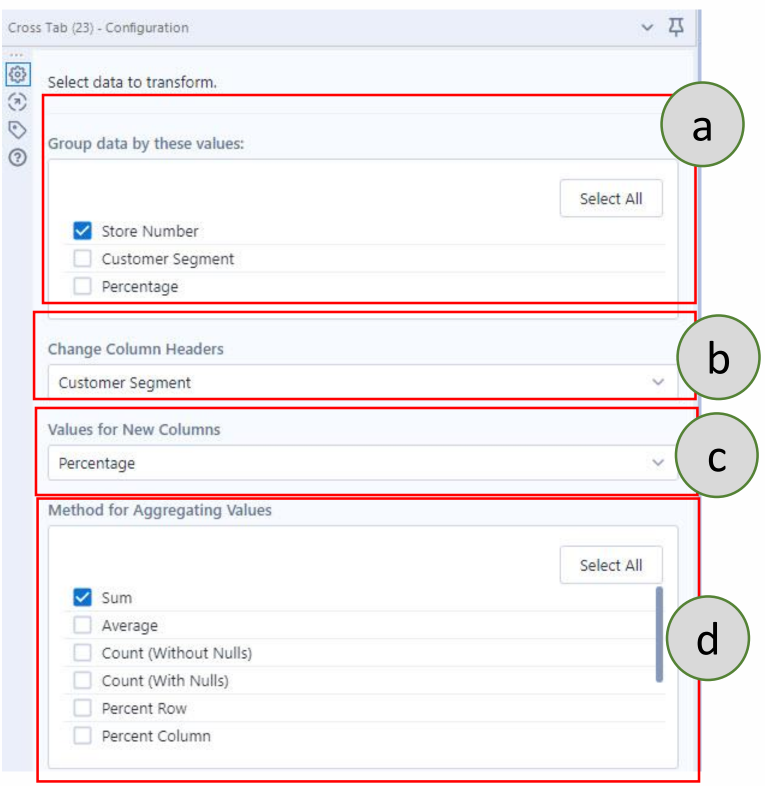
a: Select field(s) that will be key column(s).

b: Select field(s) that will be transposed as row

6.0: Transform

Cross Tab Tool

- Use **Cross Tab** tool to pivot the orientation of data in a table by moving vertical data fields onto a horizontal axis and summarizing data where specified.



a: Select field to group by the data.

b: Select field that will be used as column header after process.

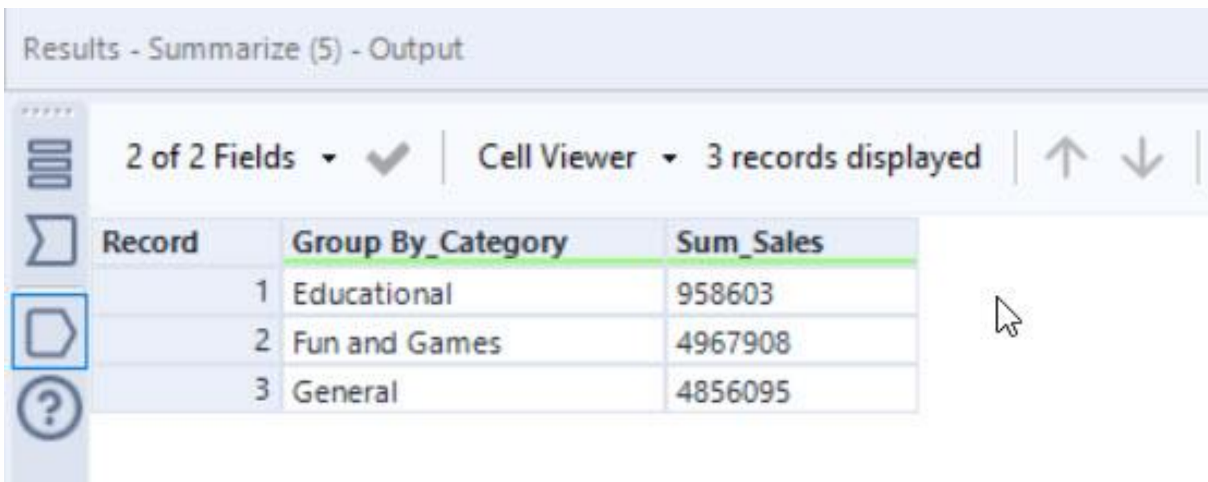
c: Select field that the values will be add on the new generated columns.

d: Select method for aggregating the value of chosen field in **c**.

Exercise #4

The data set EduFunGames.csv contains the monthly sales numbers for different categories of products. What are the total sales for each category: General, Fun and Games, and Educational?

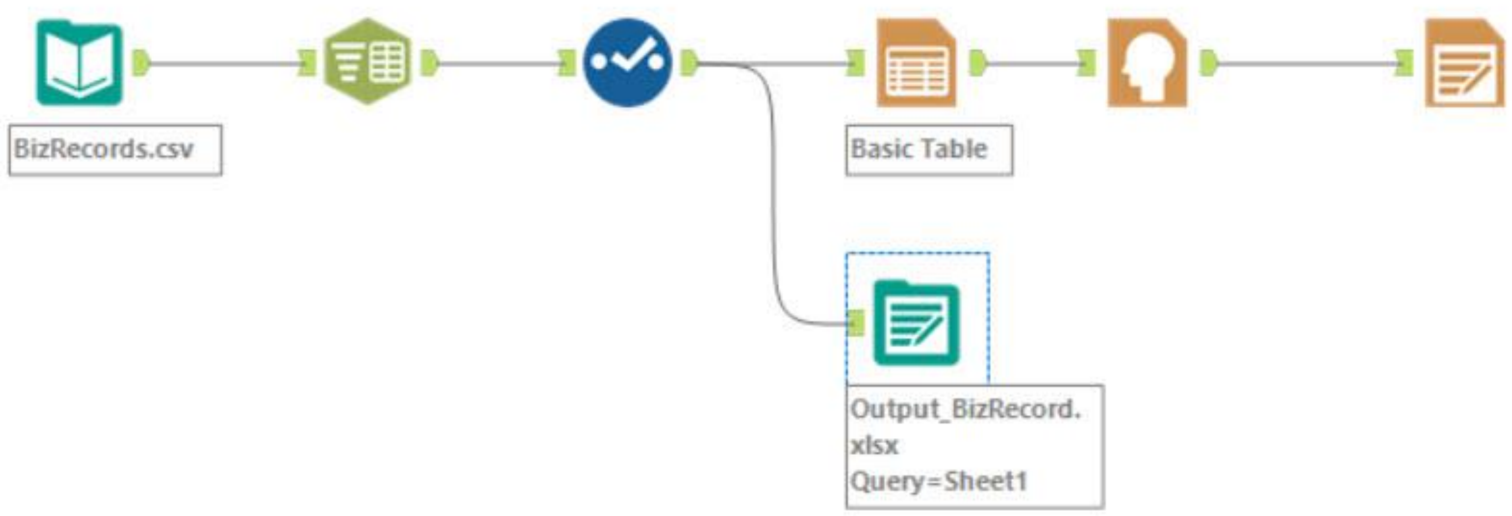
Sample solution:



Record	Group By_Category	Sum_Sales
1	Educational	958603
2	Fun and Games	4967908
3	General	4856095

7.0: Write Output Data

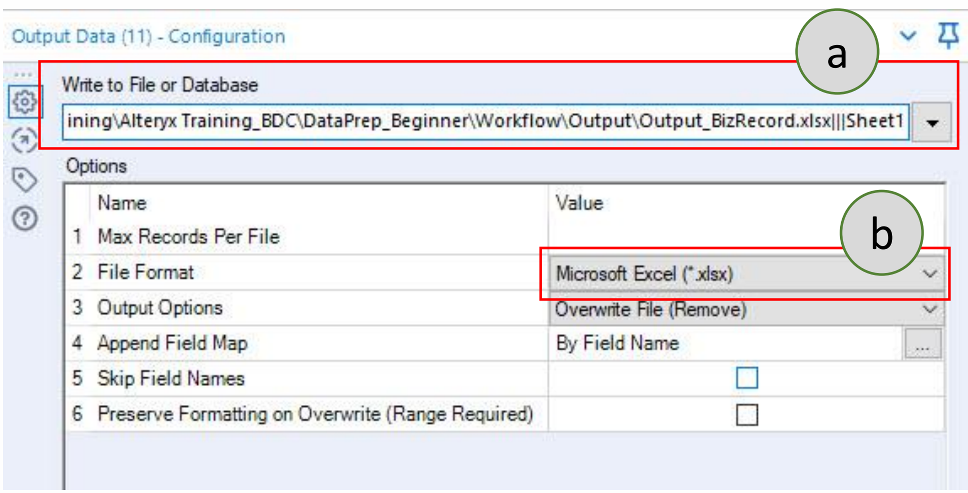
Example:

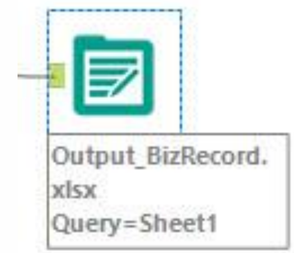


7.0: Write Output Data

Output Data Tool

- Use **Output** tool to write results of a workflow to supported file types or data sources.





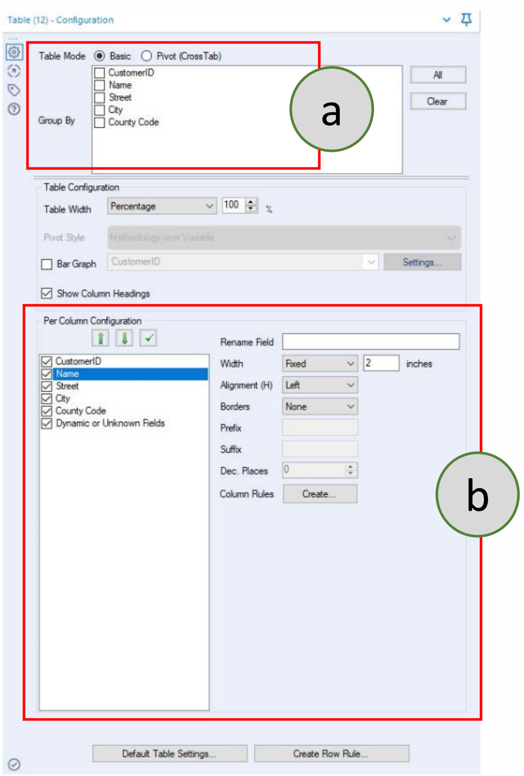
a: Setting path to save the output file.

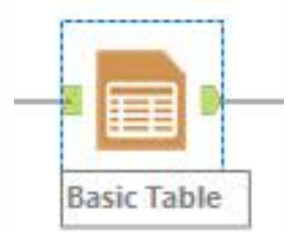
b: Check file format.

7.0: Write Output Data

Table Tool

- Use **Table** tool to create a data or pivot table to output in a report via the Render tool.





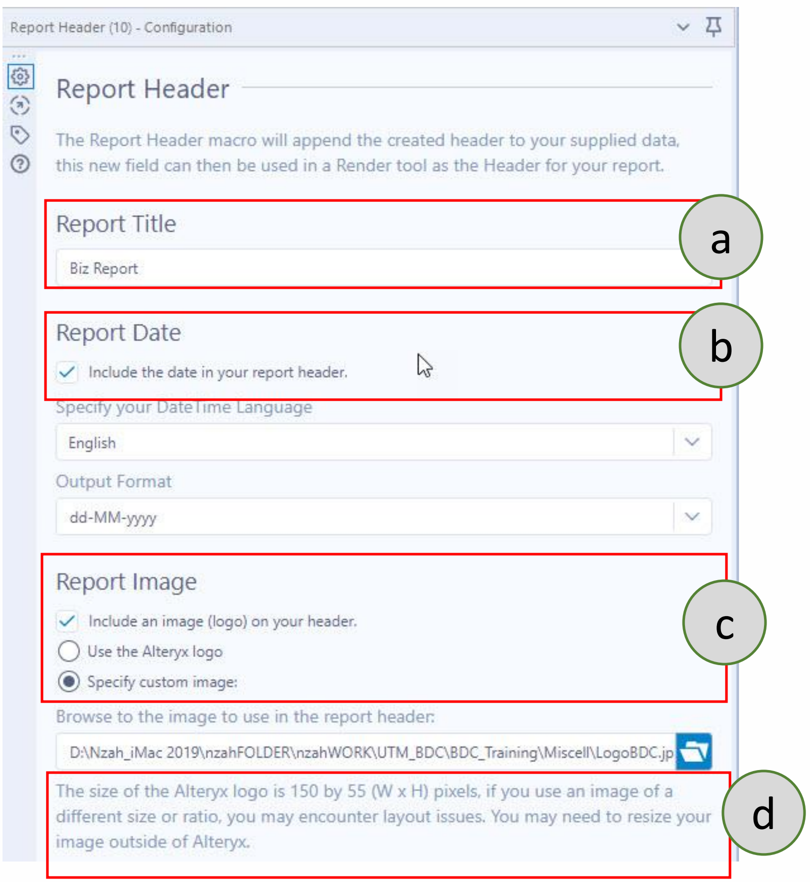
a: Select field to group by the data (optional).

b: Configure table format (optional).

7.0: Write Output Data

Report Header Tool

- Use **Report Header** tool to set up and place a header onto your report.

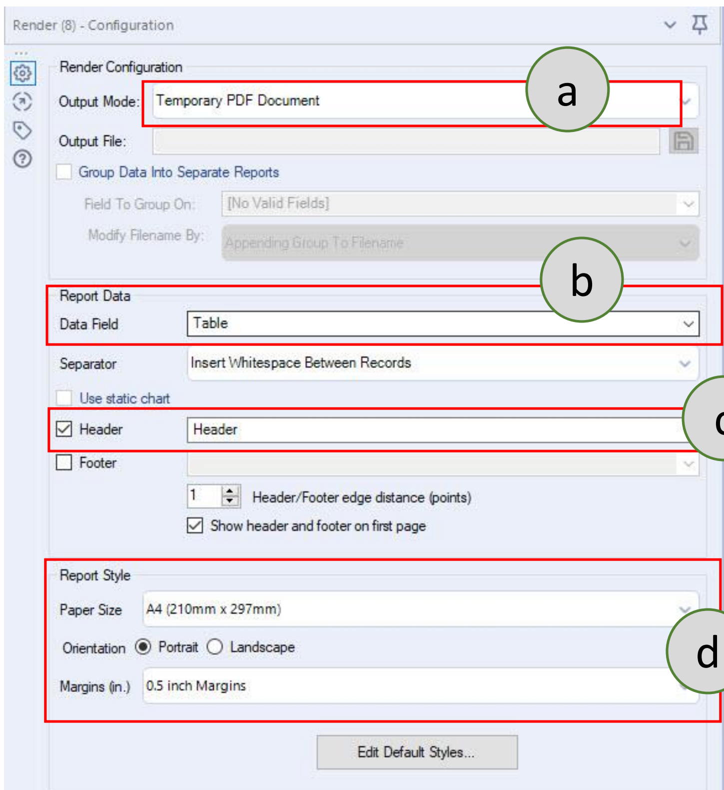


a: Set report title.
b: Tick the box to add report date.
c: Setting to add image/company logo on report.
d: Guideline of image size.

7.0: Write Output Data

Render Tool

- Use **Render** tool to transform report snippets into presentation-quality reports in a variety of formats. Output to Microsoft Office format is only supported for Office 2007 and above.



a: Select output mode/format (pdf/word/excel/..) and output file.
b: Select report data.
c: Tick Header to include a header that has been set up.
d: Configure report style.

7.0: Write Output Data

Biz Report

31-07-2021

Sample Output (pdf) – without group by data

UTM BIG DATA CENTRE

TOGETHER WE MAKE MOTIONS...

CustomerID	Name	Street	City	County Code
1	PAMELA WRIGHT	2316 E 5th Ave	Denver	CO 80206-4205
2	MELISSA RUFF	2753 S Milwaukee St	Denver	CO 80210-6426
3	CONSTANTI VLASSIS	16911 E Harvard Ave	Aurora	CO 80013-1507
4	AMY LOCKEMER	3721 S Pitkin Ct	Aurora	CO 80013-3024
5	DANELL VALDEZ	2925 W College Ave	Denver	CO 80219-6059
6	JESSICA RINEHART	4220 W 35th Ave	Denver	CO 80212-1902
7	NANCY CLARK	8785 Cloverleaf Cir	Parker	CO 80134
8	ANDREA BRUN	2080 Fenton St	Denver	CO 80214
9	DENISE PENTICO	4125C S Evanston Cir	Aurora	CO 80014
10	ERNA ARUSTAMYAN	1137 S Boston Ct	Denver	CO 80247
11	ZAREFA HAMMAD	1651 Biscay Cir	Aurora	CO 80011-5220
12	MURIEL COULTER	12704 E Asbury Cir	Aurora	CO 80014-2759
13	KAREN OSBORNE	7595 S Teller Ct	Littleton	CO 80128-5488
14	SHIRLEY KRYWONIS	370 Eudora St	Denver	CO 80220-5722
15	MICHELE MUNSEY	8300 Fairmount Dr	Denver	CO 80247-6527
16	DIANNE VANGILDER	15358 E Saratoga Pl	Aurora	CO 80015
17	NELE MENDIOLA	7498 Alkire St	Arvada	CO 80005
18	JOAN LADD	2552 E Alameda Ave	Denver	CO 80209-3303
19	DOROTHY COLE	6561 Welch Ct	Arvada	CO 80004-2223

7.0: Write Output Data

Biz Report

31-07-2021

Sample Output (pdf) - use group by Customer ID

UTM BIG DATA CENTRE
TOGETHER WE MAKE MOTIONS...

CustomerID	Name	Street	City	County Code
1	PAMELA WRIGHT	2316 E 5th Ave	Denver	CO 80206-4205

CustomerID	Name	Street	City	County Code
2	MELISSA RUFF	2753 S Milwaukee St	Denver	CO 80210-6426

CustomerID	Name	Street	City	County Code
3	CONSTANTI VLASSIS	16911 E Harvard Ave	Aurora	CO 80013-1507

CustomerID	Name	Street	City	County Code
4	AMY LOCKEMER	3721 S Pitkin Ct	Aurora	CO 80013-3024

CustomerID	Name	Street	City	County Code
5	DANELL VALDEZ	2925 W College Ave	Denver	CO 80219-6059

CustomerID	Name	Street	City	County Code
6	JESSICA RINEHART	4220 W 35th Ave	Denver	CO 80212-1902